

The sintering-timescale map (Figure 8): what goes into it

enceladus_DSM, studies/keff_sintering/timescale_map/

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Summary

Figure 8 is **not simulation output**. It is one closed-form expression,

$$t(T, R) = \theta^* \tau_{\text{sub}}(T, R), \quad (1)$$

evaluated on a grid of temperature T and grain radius R . One number comes from the campaign, the anchor $\tau_{\text{sub}} = 7822.3$ s at -20°C and $R = 50$ μm ; one result of the campaign justifies the form, the temperature collapse; everything else is the model's definition of τ_{sub} and a saturation-pressure formula. Only one grain size and temperatures from -40 to -5°C were simulated, so every other point of the map is an extrapolation. Section 8 lists what that extrapolation assumes. Every number below is recomputed by `check.timescale_map.py` in this folder (output in `check.timescale_map.txt`).

1 The phase-change time scale

The model defines (solver banner; Table S1 of the manuscript tables)

$$\tau_{\text{sub}} = \frac{\varepsilon^2 \beta_{\text{sub}}}{d_0}, \quad \beta_{\text{sub}} = \frac{\rho_{\text{ice}}}{\rho_{vs}(T)} \frac{1}{\alpha_c} \sqrt{\frac{2\pi m}{k_B T}}, \quad d_0 = \frac{\gamma a^3}{k_B T}, \quad a^3 = \frac{m}{\rho_{\text{ice}}}, \quad (2)$$

with ε the interface parameter, α_c the condensation coefficient, m the mass of a water molecule, γ the ice–air interfacial energy and ρ_{vs} the saturation vapor density over ice. Substituting,

$$\boxed{\tau_{\text{sub}} = \frac{\varepsilon^2 \rho_{\text{ice}}^2}{\alpha_c \gamma m} \frac{\sqrt{2\pi m k_B T}}{\rho_{vs}(T)}} \quad (3)$$

With the solver's own ρ_{vs} , Eq. (3) reproduces the solver's τ_{sub} at all five simulated temperatures to within 0.03% (60 389 against 60 390 s at -40°C ; 7822 against 7822 s at -20°C).

2 Temperature dependence

For water vapor as an ideal gas, $\rho_{vs} = p_{\text{sat}}(T) m / (k_B T)$, so Eq. (3) gives

$$\tau_{\text{sub}} \propto \varepsilon^2 \frac{T^{3/2}}{p_{\text{sat}}(T)}. \quad (4)$$

The map uses the Murphy and Koop (2005) expression for the saturation pressure over ice,

$$\ln p_{\text{sat}}[\text{Pa}] = 9.550426 - \frac{5723.265}{T} + 3.53068 \ln T - 0.00728332 T.$$

To verify before citing: the coefficients were entered from memory of that paper, and as recalled the fit is stated for $T > 110$ K; the map extends to 60 K.

Effective activation energy. From Eq. (4), $E = Q_{\text{sub}} - \frac{3}{2} R_g T$. With the formula above $Q_{\text{sub}} = 51.1 \text{ kJ mol}^{-1}$ and $E = 48.0 \text{ kJ mol}^{-1}$ at 253 K (50.9 and 48.7 at 180 K). This is the “about 48 kJ mol^{−1}” quoted for the vapor route.

The map and the solver do not use the same $\rho_{vs}(T)$. The solver takes

$$\rho_{vs} = 0.62 \rho_{\text{air}} \frac{P_{vs}}{P_{\text{atm}} - P_{vs}}$$

with the ASHRAE polynomial for P_{vs} and a *fixed* air density $\rho_{\text{air}} = 1.341 \text{ kg m}^{-3}$, which lacks the $1/T$ of the ideal gas law. The two agree at -10°C and drift apart on either side (Table 1). The map is anchored at -20°C , so inside the simulated range it differs from the solver’s τ_{sub} by -8% to $+6\%$. The ideal-gas form is the physically correct one; the difference is small next to the many orders of magnitude the map spans, but the map is *not* the solver’s clock evaluated elsewhere.

Table 1: The map’s τ_{sub} against the solver’s, at $R = 50 \text{ }\mu\text{m}$ ($\varepsilon = 1 \text{ }\mu\text{m}$).

T	ρ_{vs} ratio (map/solver)	τ_{sub} map (s)	τ_{sub} solver (s)	difference
-40°C	1.131	55 579	60 390	-8.0%
-30°C	1.085	20 001	20 840	-4.0%
-20°C	1.041	7 822	7 822	anchor
-10°C	1.000	3 294	3 164	$+4.1\%$
-5°C	0.980	2 192	2 063	$+6.2\%$

3 Grain-size dependence

There is **no grain-size data** in the figure. The vertical axis is the variable R in Eq. (1); only $R = 50 \text{ }\mu\text{m}$ was simulated.

The R^2 comes from the discretization rule of the campaign: the interface parameter is tied to the grain size, $\varepsilon = \bar{R}/50$, and the domain to $L = 40 \bar{R}$, so the mesh is the same at any $\bar{R}$. In Eq. (3), $\tau_{\text{sub}} \propto \varepsilon^2 = (R/50)^2$. Written with the grain radius,

$$\tau_{\text{sub}} = \frac{1}{2500} \tau_R, \quad \tau_R \equiv \frac{R^2 \beta_{\text{sub}}}{d_0}, \quad (5)$$

and τ_R is a physical time: the attachment-limited time for curvature-driven transfer over a grain radius. The numerical parameter ε has dropped out, which is why the map can be drawn against R .

This scaling is an *argument, not a result*: if the geometry is the same in units of R and attachment kinetics is the only rate-limiting step, the evolution in units of τ_R is the same. It has not been tested by running a second grain size.

4 Why time scales with τ_{sub} at all

Equation (1) assumes the microstructure is a function of the sintering age $\theta = t/\tau_{\text{sub}}$ alone. Across temperature this is the campaign’s main result (Figure 6): for each of the 25 packings, runs at five temperatures give the same k_{eff} at matched SSA to within 0.27% (100 pairs; median 0.07%), and the time needed to reach a given state follows the ratio of τ_{sub} to within 4.1%. So within -40 to -5°C the statement “time to a given state $\propto \tau_{\text{sub}}$ ” is measured, to 4%.

5 The target age θ^*

$\theta^* = 331$ is the age the -20°C runs reach in 30 days: $30 \text{ d}/7822.3 \text{ s} = 331.4$. It has no other significance; any age gives the same map shifted by a constant factor. In the results, at $\theta = 331$ the 15 well-connected packings (porosity ≤ 0.375) have

$$k_{\text{eff}}/k_{\text{eff},r} = 1.171 \pm 0.015, \quad \text{SSA}/\text{SSA}_r = 0.819 \pm 0.007$$

relative to the reference age $\theta_r = 30$ (mean and standard deviation over packings). The map is therefore “time for k_{eff} to rise 17% and SSA to fall 18% from their values at θ_r ”.

6 The formula as plotted

$$t(T, R) = \theta^* \tau_{\text{sub}}^{\text{ref}} \frac{T^{3/2}/p_{\text{sat}}(T)}{T_{\text{ref}}^{3/2}/p_{\text{sat}}(T_{\text{ref}})} \left(\frac{R}{R_{\text{ref}}} \right)^2, \quad (6)$$

with $\theta^* = 331$, $\tau_{\text{sub}}^{\text{ref}} = 7822.3 \text{ s}$, $T_{\text{ref}} = 253.15 \text{ K}$ and $R_{\text{ref}} = 50 \mu\text{m}$. It is evaluated for $60 \leq T \leq 260 \text{ K}$ and $0.1 \mu\text{m} \leq R \leq 1 \text{ mm}$ and binned at 1 h, 1 d, 1 yr, 1 kyr, 1 Myr and 4.5 Gyr (`fig.timescales` in `figures/sample_figures.py`).

Table 2: Values of the map. Only the first row, last column, is a simulated condition.

T (K)	$R = 1 \mu\text{m}$	$5 \mu\text{m}$	$6 \mu\text{m}$	$50 \mu\text{m}$
253.15	17 min	7.2 h	10.4 h	30 d
220	9.1 h	9.4 d	13.6 d	2.6 yr
200	5.3 d	134 d	192 d	37 yr
180	137 d	9.4 yr	13.6 yr	941 yr
150	253 yr	6.3 kyr	9.1 kyr	633 kyr
120	4.4 Myr	111 Myr	160 Myr	11 Gyr
110	377 Myr	9.4 Gyr	13.6 Gyr	> age of the Solar System
80		> 10^{17} yr at every grain size		

7 The annotations and their sources

The three unsourced brackets need a reference each or should be removed.

Annotation	Value	Source
“this study”	253.15 K, 50 μm	the campaign
Choukroun et al. (2020)	180 K, 6 μm , 15 yr	their paper (read for this study): 12 μm diameter grains; about 15 yr to a strength of 10 MPa at 180 K
“plume grains” bracket	0.1–5 μm	none: entered from general recollection of the Enceladus literature
“surface” bracket	60–80 K	none: same
“fractures” bracket	175–185 K	none: same

8 What the extrapolation assumes

1. **Temperature range.** Simulated: 233–268 K. Plotted: 60–260 K. The collapse is measured over 35 K and applied over 200 K.
2. **One grain size.** The R^2 law is a scaling argument (Section 3).
3. **Attachment-limited transfer everywhere.** The crossover length between attachment and diffusion control is $L^* = D_v \beta_{HK}$, $\beta_{HK} = \alpha_c^{-1} \sqrt{2\pi m / (k_B T)}$: 139 μm at 253 K and 89 μm at 180 K for air at 1 atm and $\alpha_c = 10^{-3}$. The R^2 law needs $R \ll L^*$. The simulated $R = 50 \mu\text{m}$ is already at $R/L^* = 0.36$, and the upper third of the map ($R > 100 \mu\text{m}$) is beyond it; there diffusion slows growth and the map *underestimates* the time.
4. **Constant $\alpha_c = 10^{-3}$** at all temperatures. Its value near 100 K is not known to us.
5. **Air-filled pores at 1 atm.** The pore gas enters through D_v and L^* . Near the surface of Enceladus the pores are close to vacuum.
6. **Vapor route only.** Surface diffusion is absent from the model. Molaro et al. (2019) and Choukroun et al. (2020) both attribute cold sintering to it, the latter with $Q = 24.3 \pm 3.3 \text{ kJ mol}^{-1}$, half of our 48. A missing mechanism can only make sintering faster, so the map is an *upper bound on the time* taken by the vapor route alone.
7. **The saturation-pressure fit** is used below its stated range (to be verified, Section 2).
8. **The age–state relation** (SSA linear in $\log \theta$) is itself an intermediate-time approximation; the map extrapolates the clock, not that relation.

The Choukroun point. The map gives 13.6 yr at 180 K and $R = 6 \mu\text{m}$; Choukroun et al. give about 15 yr. These are not the same quantity. Theirs is the time to a strength of 10 MPa by a mechanism they identify as surface diffusion; ours is the time to an arbitrary sintering age by vapor transport. The closeness of the two numbers is not evidence that the model reproduces their experiment.

9 Reproducing it

cd studies/keff_sintering

```
python timescale_map/check_timescale_map.py <campaign dir>  
python figures/sample_figures.py <campaign dir> --skip-gallery  
# the second writes Figure8_timescales; use venv_enceladus/bin/python
```